

# Ricardo Andrés Calvo Méndez

Ph.D. Student in Electrical and Computer Engineering  
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## SUMMARY

Ph.D. student in Electrical and Computer Engineering at Purdue University working on **formal verification**, **program analysis**, **software security**, and **performance optimization**. My research automates program analysis for vulnerability discovery and optimization opportunities. I bring **3+ years of industry experience** in full-stack development and quality assurance, including leading development teams.

**Research interests:** formal verification; program analysis and testing; performance optimization.

## EDUCATION

**Ph.D. in Electrical and Computer Engineering** Aug 2025 – Expected May 2030  
*Purdue University, West Lafayette, IN*

**B.Sc. in Systems and Computer Engineering** Feb 2018 – Oct 2023  
*Universidad Nacional de Colombia (ranked #1 in Colombia), Bogotá, Colombia. Class rank: 2/49.*

## RESEARCH EXPERIENCE

**Graduate Research Assistant** Aug 2025 – Present  
*Purdue University — Advisor: James C. Davis*

- Co-develop *AutoSOUP*, a system that combines large language models with the CBMC bounded model checker to automate memory-safety verification of C components (ACM CCS 2026); contributed to system design.
- Designed AutoSOUP's **debugging agent**, which reads verification errors, diagnoses why a proof fails, and proposes a fix.
- Built the experimental setup used to run AutoSOUP's large-scale evaluation on a computing cluster.
- Across **177 components** from four embedded operating systems, AutoSOUP exposed **66.7% of 60 known CVEs** (vs. 28.3% for a frontier coding agent) and uncovered **20 new vulnerabilities** reported to maintainers.
- Co-authored studies on false-positive crashes in Google's OSS-Fuzz-Gen (FSE 2026) and on how AI agents optimize code (MSR 2026); presenting author of both.

**Undergraduate Visiting Scholar (UREP-C)** Feb 2023 – Jul 2023  
*Purdue University — Advisor: James C. Davis*

- With mentorship, designed and implemented the fuzzing strategy of *EmNetTest*, a packet-validation testing framework, and its evaluation on **4 embedded network stacks**, using C, C++ and Python.
- Contributed to the manual triage of results; EmNetTest reproduced **12 known vulnerabilities** and found **7 new defects (6 CVEs)**, leading to an ASE 2023 publication.

## PROFESSIONAL EXPERIENCE

**Test Engineer & QA Analyst** Sep 2024 – Jul 2025  
*Kravata S.A.S., Bogotá, Colombia*

- Joined as the company's **first and only QA engineer** and defined its testing strategy, standards, and processes from scratch.
- Led testing for the company's **4 most critical services**.
- Automated unit, integration and end-to-end tests with Python, Cucumber, Selenium and Flask, and integrated them into CI/CD pipelines.

## Full-Stack Software Developer

Aug 2023 – Aug 2024

*Fit Ideas S.A.S., Bogotá, Colombia*

- Promoted to team lead; led **4 developers** building and deploying a **white-label** cross-platform fuel sales application, customizable per client brand.
- Defined the system architecture, ran code reviews, and supported the team's Scrum process.
- Built web and mobile systems with Django, React, Spring Boot, Flutter and PostgreSQL, including end-to-end tests.

## Junior Software Developer

Aug 2021 – Jan 2023

*Fit Ideas S.A.S., Bogotá, Colombia*

- Built a configurable web scraper that collects public online data on demand.
- Developed web and mobile product features with Django, React, Spring Boot and Flutter, and designed and maintained PostgreSQL databases.

## TEACHING EXPERIENCE

### Graduate Teaching Assistant

Fall 2026

*ECE 30861: Software Engineering — Purdue University*

- Grade assignments and hold office hours for a course of ~150 students.

## TECHNICAL SKILLS

**Research & Security:** Formal verification, memory-safety verification, bounded model checking (CBMC), program analysis, fuzzing, performance optimization

**Programming:** C, C++, Python, Java, JavaScript, TypeScript, Dart, SQL

**AI for SE:** LLM agents, agentic pipelines, empirical evaluation

**Engineering:** Spring Boot, React, Django, Flask, Flutter, PostgreSQL, Selenium, Cucumber, CI/CD, Docker, Linux, Git

## REFEREED CONFERENCE PUBLICATIONS

- [C-1] P.C. Amusuo, R. Calvo, D. Anandayuvraj, T. Le Lievre, K. Kolyakov, E. Jorgensen, A. Machiry, and J.C. Davis. *AutoSOUP: Safety-Oriented Unit Proof Generation for Component-level Memory-Safety Verification* *arXiv*. Accepted at the 33rd ACM Conference on Computer and Communications Security (CCS'26). *Presenting author*.
- [C-2] P. Amusuo, D. Liu, R. Calvo, J. Metzman, O. Chang, and J.C. Davis. *Lessons from Mitigating False Positives in Google's OSS-Fuzz-Gen* *DOI*. Proceedings of the ACM International Conference on the Foundations of Software Engineering (FSE-Industry'26). *Presenting author*.
- [C-3] H. Peng, A. Z. Qiu, R. Calvo, K.G. Kalu, and J.C. Davis. *How Do Agents Perform Code Optimization? An Empirical Study* *DOI*. Proceedings of the 23rd International Conference on Mining Software Repositories — Mining Challenge track (MSR-Mining'26). *Presenting author*.
- [C-4] P.C. Amusuo, R. Calvo, Z. Xu, A. Machiry, and J.C. Davis. *Systematically Detecting Packet Validation Vulnerabilities in Embedded Network Stacks* *DOI*. Proceedings of the 38th IEEE/ACM International Conference on Automated Software Engineering (ASE'23).

## PRELIMINARY WORKS

- [P-1] P.C. Amusuo, D. Liu, R. Calvo, J. Metzman, O. Chang, and J.C. Davis. *FalseCrashReducer: Mitigating False Positive Crashes in OSS-Fuzz-Gen Using Agentic AI* *arXiv*. Technical preprint (2025).

## ACADEMIC SERVICE

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|----------------------------------------------|------|
| Reviewer — MSR'26 Mining Challenge Track     | 2026 |
| Sub-reviewer — ICSE'27                       | 2026 |
| Sub-reviewer — ICSE'26                       | 2026 |
| Sub-reviewer — USENIX Security '26 (Cycle 2) | 2026 |
| Student Volunteer — ICSE'26                  | 2026 |

## GRANTS AND SCHOLARSHIPS

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| <b>CCS'26 Student Conference Grant</b>                                                          | 2026 |
| <i>ACM CCS</i>                                                                                  |      |
| <i>Travel support for CCS in The Hague, Netherlands.</i>                                        |      |
| <b>ICSE'26 Student Mentoring Workshop (SMeW)</b>                                                | 2026 |
| <i>National Science Foundation (NSF) and ACM SIGSOFT</i>                                        |      |
| <i>Travel support for the Student Mentoring Workshop in Rio de Janeiro, Brazil.</i>             |      |
| <b>“Ser Pilo Paga” Scholarship</b>                                                              | 2017 |
| <i>Ministry of Education of Colombia</i>                                                        |      |
| <i>Merit scholarship for a 99th-percentile national exam score; full undergraduate tuition.</i> |      |